GEOS-MPM Examples Suite Report

Generated by examples/runExampleSuite

May 17, 2026

Running and rerunning

Each example directory contains a copyable `pfw_input_*.py` and a `runProblem` wrapper. The wrapper stages an isolated case directory under `testRunDirectory`, copies the needed PFW files, runs the particle-file writer, submits GEOS, submits dependent post-processing, and copies figures/logs back to the example `output/` directory.

```
cd scripts/preProcessing/materialPointMethodParticleFileWriter/examples
./runExampleSuite all --python /usr/tce/bin/python3 --force
./runExampleSuite all --case borehole_Ghareb --python /usr/tce/bin/python3 --force
./runExampleSuite report --python /usr/tce/bin/python3
```

If output exists, `runProblem` prompts before overwriting. Use `--force` for noninteractive reruns. The suite records case-level failures, continues with remaining cases by default, and reports incomplete or interrupted Slurm jobs explicitly.

Timing and completion summary

Example	Status	GEOS state	GEOS wall	GEOS CPU	Post state	Post wall
brazilianDisk_FMPM	figures available					
brazilianDisk_BSpline	figures available					
brazilianDisk_FLIP	figures available					
brazilianDisk_PIC	figures available					
brazilianDisk_XPIC	figures available					
benchy	figures available					
borehole_Ghareb	post FAILED	COMPLETED	00:00:23	00:00:00	FAILED	00:00:02
collidingDisks	post FAILED	COMPLETED	00:00:06	00:00:00	FAILED	00:00:02
elasticDisk	post FAILED	COMPLETED	00:00:49	00:00:00	FAILED	00:00:00
pdc	post FAILED	COMPLETED	00:09:55	00:00:00	FAILED	00:00:01

Incomplete, failed, or missing-output cases

borehole_Ghareb – post FAILED

GEOS job/state: 6046259 / COMPLETED
Post job/state: 6046260 / FAILED No diagnostic log excerpt was found. The case may not have started or may still be queued/running.

collidingDisks – post FAILED

GEOS job/state: 6046261 / COMPLETED
Post job/state: 6046262 / FAILED No diagnostic log excerpt was found. The case may not have started or may still be queued/running.

elasticDisk – post FAILED

GEOS job/state: 6046263 / COMPLETED
Post job/state: 6046264 / FAILED No diagnostic log excerpt was found. The case may not have started or may still be queued/running.

pdc – post FAILED

GEOS job/state: 6046265 / COMPLETED
Post job/state: 6046266 / FAILED No diagnostic log excerpt was found. The case may not have started or may still be queued/running.

brazilianDisk_FMPM

Plane-strain Brazilian disk compression with quartz damage, Weibull strength scaling, and CPDI particles. **Status:** figures available

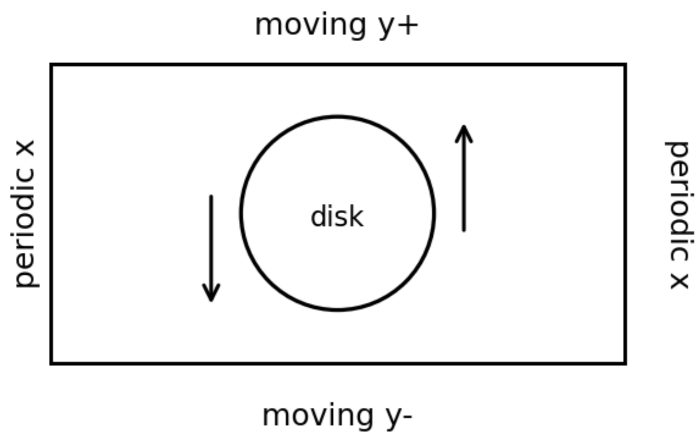

Figure 1: Schematic for brazilianDisk_FMPM.

Code features exercised.

- FMPM update
- CPDI particles
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

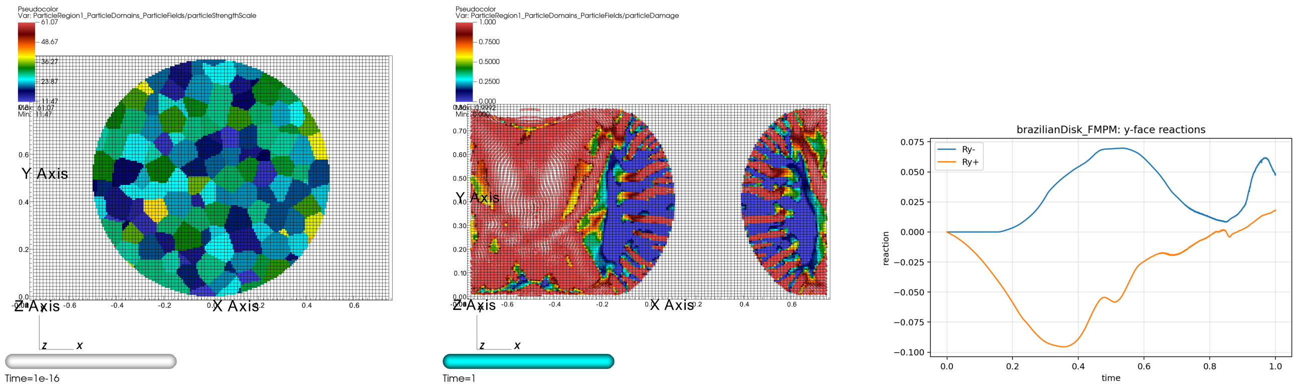

Figure 2: Initial state, final state, and reaction history for brazilianDisk_FMPM.

Output figures: brazilianDisk_FMPM_final_particleDamage.png, brazilianDisk_FMPM_initial_particleStrengthScale.png, brazilianDisk_FMPM_reactions_y.png

brazilianDisk_BSpline

Brazilian disk compression using the SinglePointBSpline particle type. The case keeps the FMPM transfer path and quartz damage material while exercising the new particle interpolation option. **Status:** figures available

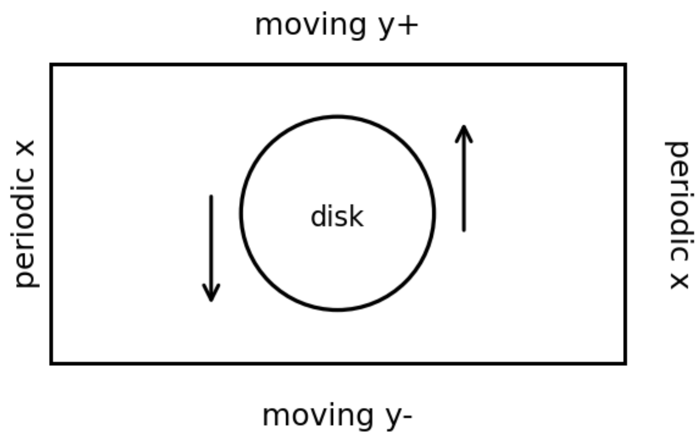

Figure 3: Schematic for brazilianDisk_BSpline.

Code features exercised.

- SinglePointBSpline particle type
- FMPM update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

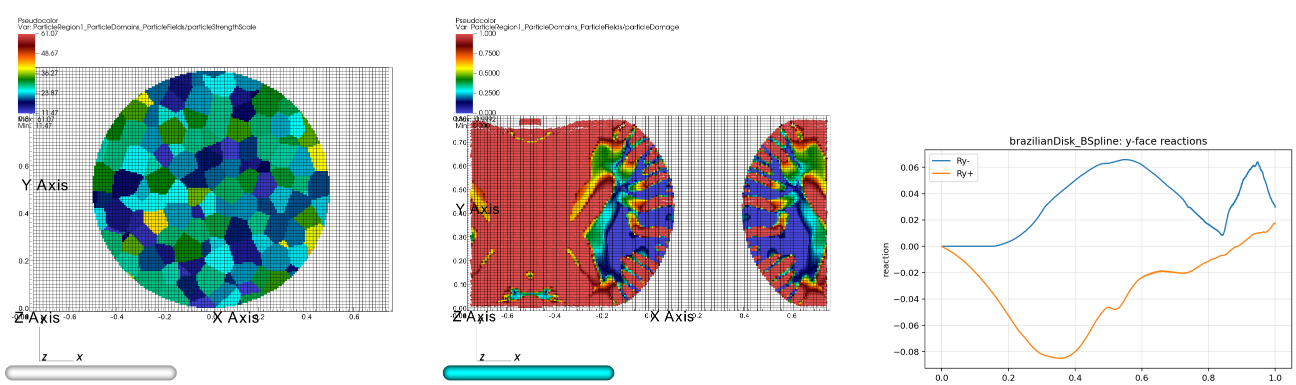

Figure 4: Initial state, final state, and reaction history for brazilianDisk_BSpline.

Output figures: brazilianDisk_BSpline_final_particleDamage.png, brazilianDisk_BSpline_initial_particleStrengthScale.png, brazilianDisk_BSpline_reactions_y.png

brazilianDisk_FLIP

Brazilian disk variant using FLIP grid-to-particle velocity transfer. **Status:** figures available

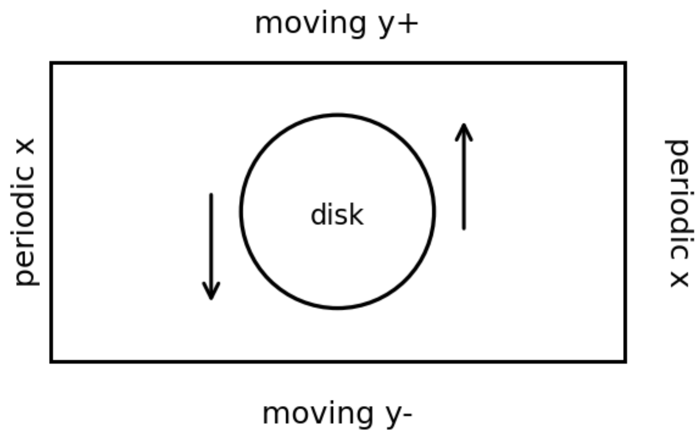

Figure 5: Schematic for brazilianDisk_FLIP.

Code features exercised.

- FLIP update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

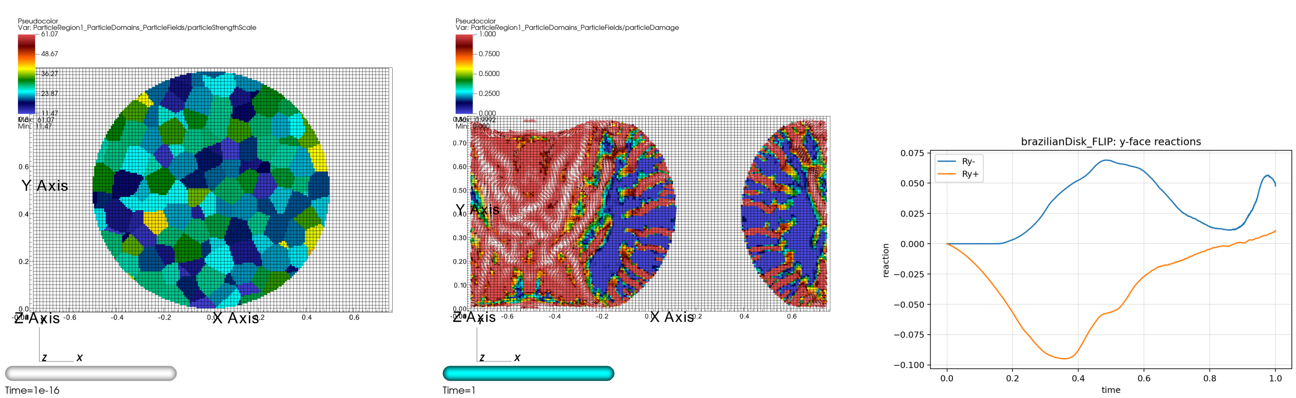

Figure 6: Initial state, final state, and reaction history for brazilianDisk_FLIP.

Output figures: brazilianDisk_FLIP_final_particleDamage.png, brazilianDisk_FLIP_initial_particleStrengthScale.png, brazilianDisk_FLIP_reactions_y.png

brazilianDisk_PIC

Brazilian disk variant using PIC velocity transfer. **Status:** figures available

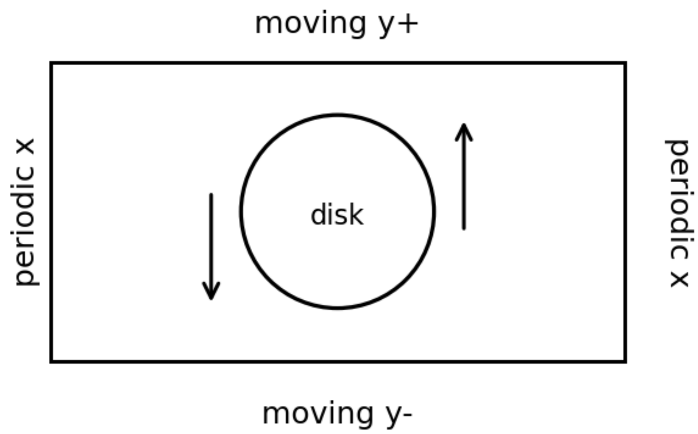

Figure 7: Schematic for brazilianDisk_PIC.

Code features exercised.

- PIC update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

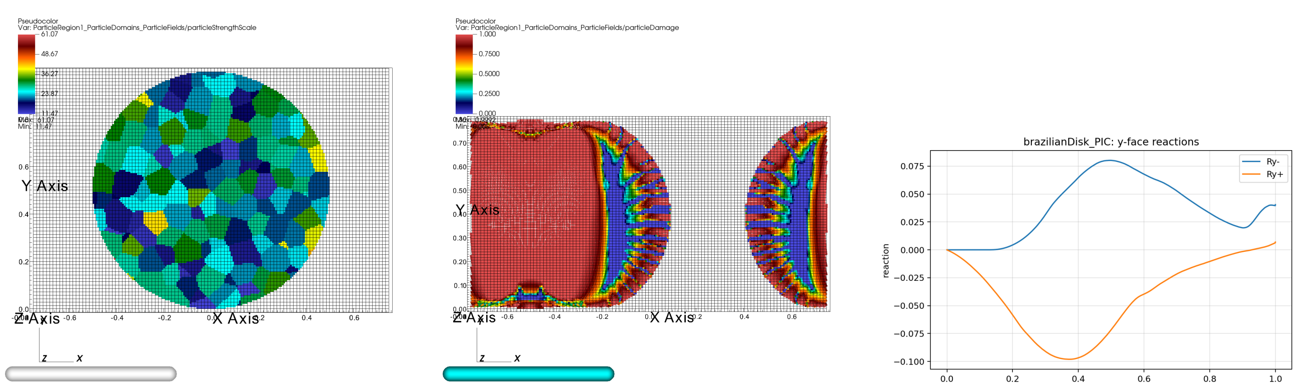

Figure 8: Initial state, final state, and reaction history for brazilianDisk_PIC.

Output figures: brazilianDisk_PIC_final_particleDamage.png, brazilianDisk_PIC_initial_particleStrengthScale.png, brazilianDisk_PIC_reactions_y.png

brazilianDisk_XPIC

Brazilian disk variant using XPIC velocity transfer. **Status:** figures available

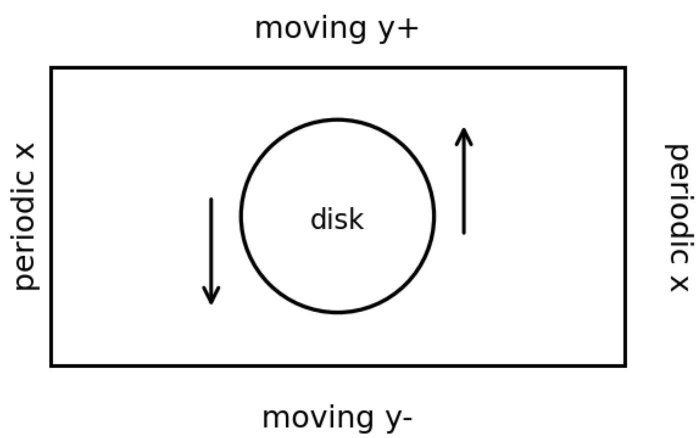

Figure 9: Schematic for brazilianDisk_XPIC.

Code features exercised.

- XPIC update
- quartz damage material
- moving y boundaries
- periodic x boundaries
- reaction history

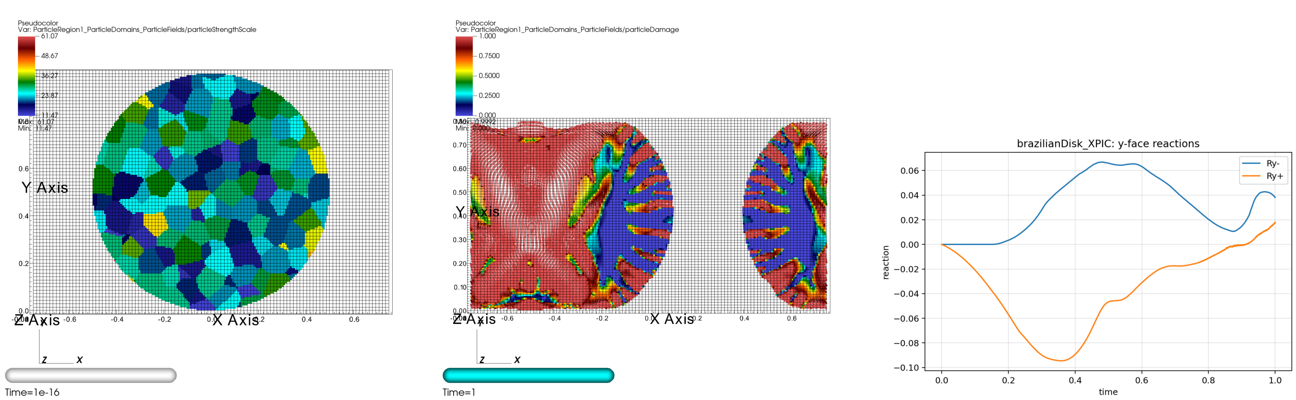

Figure 10: Initial state, final state, and reaction history for brazilianDisk_XPIC.

Output figures: brazilianDisk_XPIC_final_particleDamage.png, brazilianDisk_XPIC_initial_particleStrengthScale.png, brazilianDisk_XPIC_reactions_y.png

benchy

STL import and prescribed multi-field contact between a copper 3DBenchy boat and a steel ball. **Status:** figures available

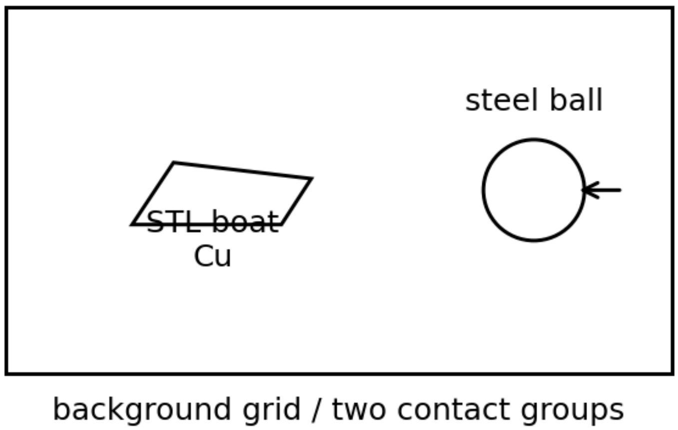

Figure 11: Schematic for benchy.

Code features exercised.

- STL geometry import
- copper boat and steel ball
- separate particle regions/contact groups
- prescribed multi-field contact
- multi-region VisIt rendering

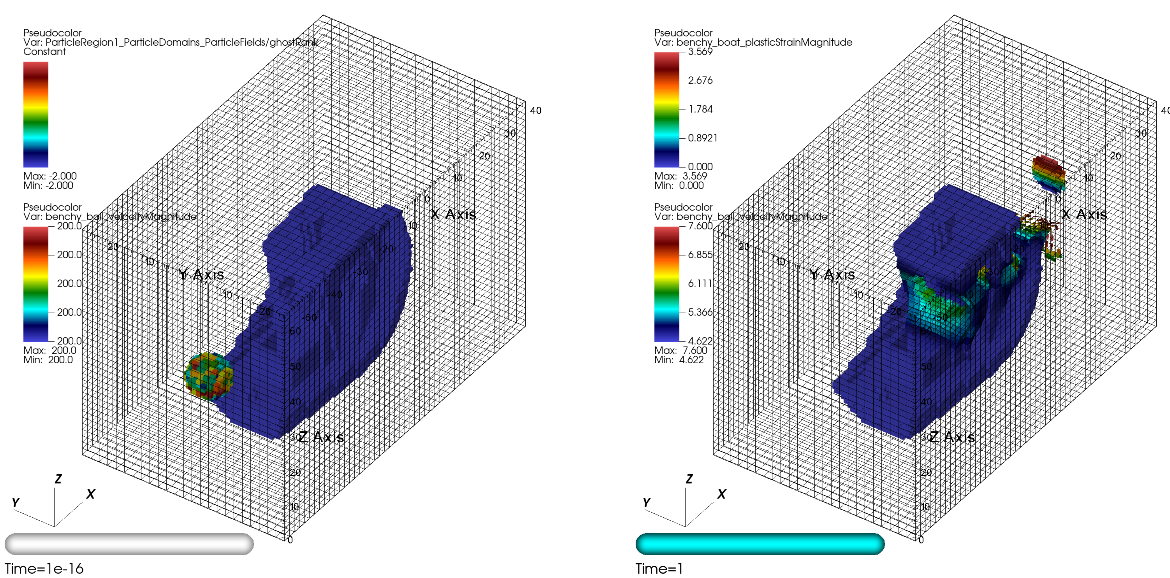

Figure 12: Initial and final state for benchy.

Output figures: benchy_FMPM_final_boatPlasticStrainMagnitude_ballVelocityMagnitude.png, benchy_FMPM_initial_boatGhostI

borehole_Ghareb

A plane-strain borehole collapse demonstration using the Ghareb geomechanics parameterization from pfw_materials.py. The rendered initial state shows hydrostatic/mean pressure and the final state shows plastic-strain magnitude. **Status:** post FAILED

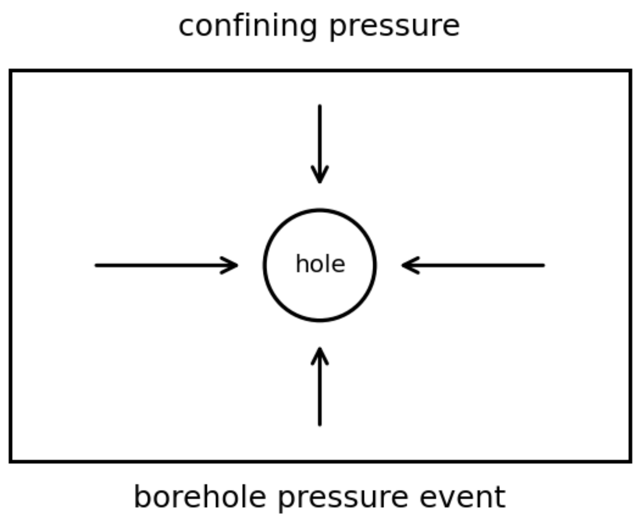

Figure 13: Schematic for borehole_Ghareb.

Code features exercised.

- Geomechanics material model
- InitializeStress MPM event
- confining-pressure event
- borehole-pressure event
- Voronoi-Weibull flaw field
- plane-strain CPDI

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Figure 14: Initial and final state for borehole_Ghareb.

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collidingDisks

Two disk-shaped particle bodies move toward one another in plane strain. The case exercises moving material points, contact-like interactions, and the XPIC update path. **Status:** post FAILED

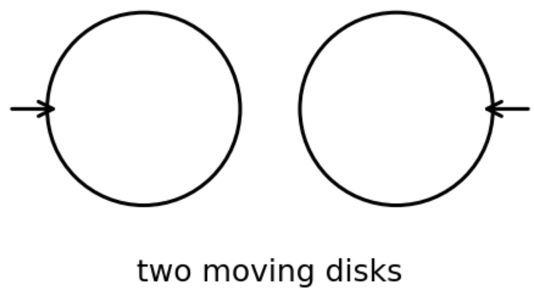

Figure 15: Schematic for collidingDisks.

Code features exercised.

- cylinder disk geometry
- XPIC update
- multi-object particle initialization
- box-average or reaction-style output when enabled

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Figure 16: Initial and final state for collidingDisks.

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elasticDisk

An elastic disk compressed by prescribed boundary motion. It is a compact example of F-table interpolation, reaction history, and elastic response. **Status:** post FAILED

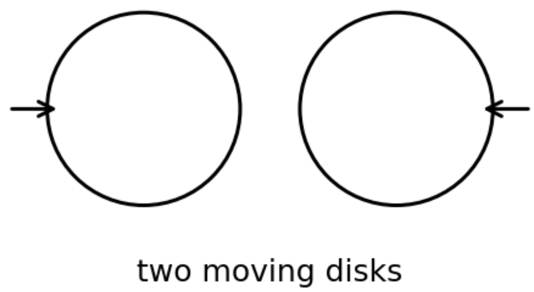

Figure 17: Schematic for elasticDisk.

Code features exercised.

- ElasticIsotropic material
- cosine F-table boundary motion
- reaction-history output when enabled
- plane-strain CPDI

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Figure 18: Initial and final state for elasticDisk.

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pdc

A small PDC cutter example with a cutter interacting with a substrate. The final visualization uses the stored three-dimensional VisIt GUI view and overlays the background grid with plastic-strain-magnitude pseudocolors for both particle regions. **Status:** post FAILED

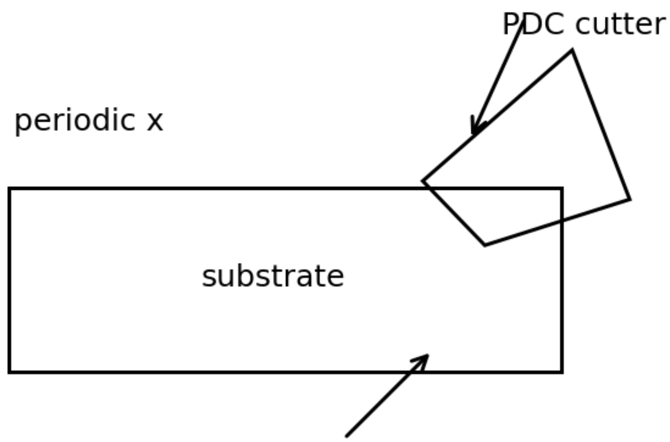

Figure 19: Schematic for pdc.

Code features exercised.

- periodic x boundary
- prescribed transverse boundary velocity
- PDC cutter/substrate geometry
- implicit contact gap correction
- two-particle-region plastic strain visualization
- stored 3D VisIt view

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Figure 20: Initial and final state for pdc.

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